HIE Knowledge Retrieval Engine

Detailed Technical Proposal

Executive Summary

The HIE Knowledge Retrieval Engine is a scalable, enterprise-grade healthcare data platform designed for X Company to solve the challenge of fragmented Electronic Health Record (EHR) data. The platform combines a high-throughput Semantic ETL backend with a Retrieval-Augmented Generation (RAG) intelligence layer to deliver accurate, real-time clinical insights through a secure natural-language interface. The system abstracts the complexity of ingesting, validating, normalizing, and enriching healthcare data at scale while ensuring compliance, privacy, and operational reliability.

1. System Architecture Overview

The system is architected as a modular pipeline consisting of data ingestion, orchestration, transformation, delivery, and intelligence layers. Each layer is independently scalable and fault-tolerant, enabling high-volume processing of clinical data across multiple healthcare partners.

2. Semantic ETL Backend (HIE Alerts Orchestration Engine)

2.1 Data Ingestion Layer

The ingestion layer supports both real-time and batch-based data acquisition from heterogeneous healthcare systems. API polling connectors integrate with EHR platforms such as Elation and Athena to retrieve patient demographics, encounters, and clinical history. Secure SFTP batch processing enables large-scale file transfers from hospital and payer partners. Direct SQL connectors provide controlled access to HIE clinical databases for structured data extraction. The engine natively supports healthcare formats including CSV, JSON, XML, HL7 v2.x, and CCDA.

2.2 Orchestration and Processing

A custom orchestration framework built on scheduled jobs and distributed file-locking mechanisms ensures safe concurrent execution without race conditions. Large datasets are handled using intelligent CSV splitting and XML chunking strategies to prevent memory exhaustion. The processing layer applies automated de-identification for compliance, duplicate detection using hashing algorithms, and configurable business rules to inject or correct missing payer and insurance information.

2.3 Transport and Output Channels

Processed and validated data is delivered through multiple enterprise-grade channels to ensure interoperability. This includes HL7 messaging via Mirth Connect, encrypted archival and staging in AWS S3 and Azure Blob Storage, CRM synchronization with Salesforce Health Cloud, write-back to SQL data warehouses, secure communications via SMTP, SMS, and Direct Secure Messaging (HISP/SES), managed file transfer platforms such as Box and Files.com, and direct partner integrations including Tricare, CrossTX, Connective Health, and Watershed. Centralized logging and monitoring are provided through Kibana.

3. Clinical Intelligence and RAG Layer

Once sanitized by the ETL engine, clinical data is transformed into a knowledge base optimized for semantic retrieval. CCDA documents are parsed into structured JSON, and related clinical observations are semantically grouped into sections such as vitals, labs, medications, and encounters using standardized vocabularies including LOINC and SNOMED. Clinical text is embedded into a ChromaDB vector store, enabling high-precision semantic search. Local LLM inference using Llama 3.2 via Ollama generates concise, context-grounded clinical summaries while ensuring patient data never leaves the environment.

4. Business Impact and Outcomes

The HIE Knowledge Retrieval Engine delivers measurable operational and clinical value. Automation of ETL workflows reduces manual data handling and error correction by approximately 80 percent, allowing engineering teams to focus on higher-value logic and integration work. Clinicians can retrieve patient-specific insights in seconds rather than spending 15–20 minutes on manual chart review. Robust orchestration, validation, and locking mechanisms ensure data integrity and zero data loss during high-volume synchronizations.

5. Technology Stack

Category	Technologies
Languages	Python 3.6.8 (Legacy Core), SQL
ETL & Orchestration	Pandas, APScheduler, Paramiko, Boto3
Integrations	Mirth Connect, Salesforce Health Cloud, Tricare API, CrossTX
Infrastructure	Docker, AWS Secrets Manager, Linux, File Locking
AI & Storage	ChromaDB, Ollama (Llama 3.2), Gradio UI